

Why vessels, and what we asked

Breast tumours are treated with chemotherapy *before* surgery. Patients who reach **pCR** — pathologic complete response, no residual invasive cancer at surgery — do markedly better, and knowing early who will get there would let clinicians change course while it still matters.

Tumours recruit their own blood supply, and **DCE-MRI** — dynamic contrast-enhanced MRI — images the breast repeatedly while a contrast agent washes through it. We ask whether the vessel network, not the tumour, carries the response signal.

Is there signal in the vascular networks?

1 Is there signal in the blood vessels at all?

2 Does the graph add anything beyond that?

The two answers stay separate. Collapsing them into one “our model works” claim would hide which half is doing the work.

From scan to prediction

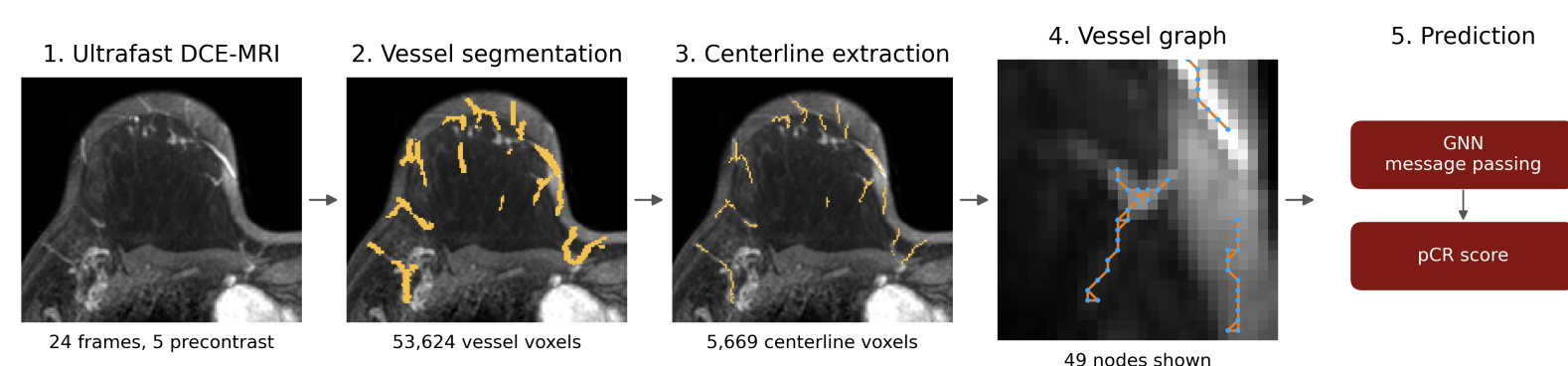
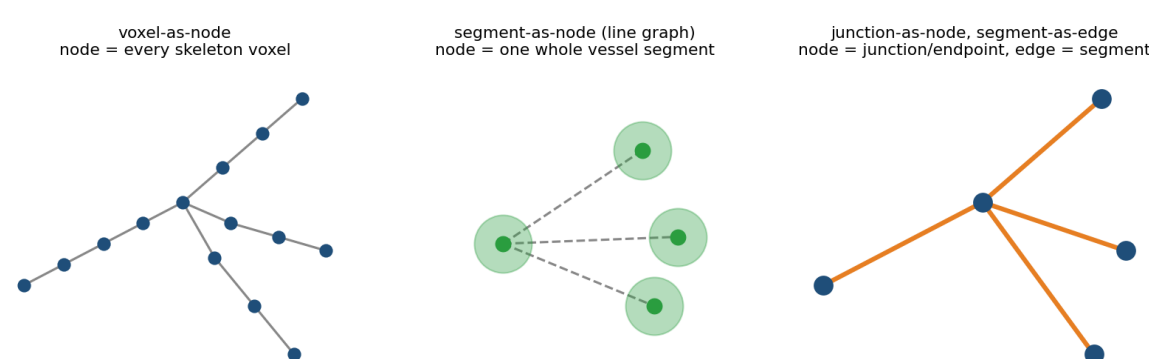


Fig. 1: Ultrafast DCE-MRI to vessel graph to prediction, on one real case. Panels 1–4 are measured data; panel 5 is schematic.

One real case: 24 frames over 119.5 s, 5 of them before contrast arrives. A **centerline** is the one-voxel-wide skeleton down the middle of each segmented vessel; every centerline voxel becomes a graph node. Segmentation and centerline extraction are upstream lab pipeline.

What the graph is

Nodes. Three representations, built from the same skeleton and compared on identical folds: one node per *centerline voxel*, per *vessel segment*, or at *branch points and tips*.



Edges. Voxel adjacency, a shared branch point, or the segments themselves, kept as a multigraph.

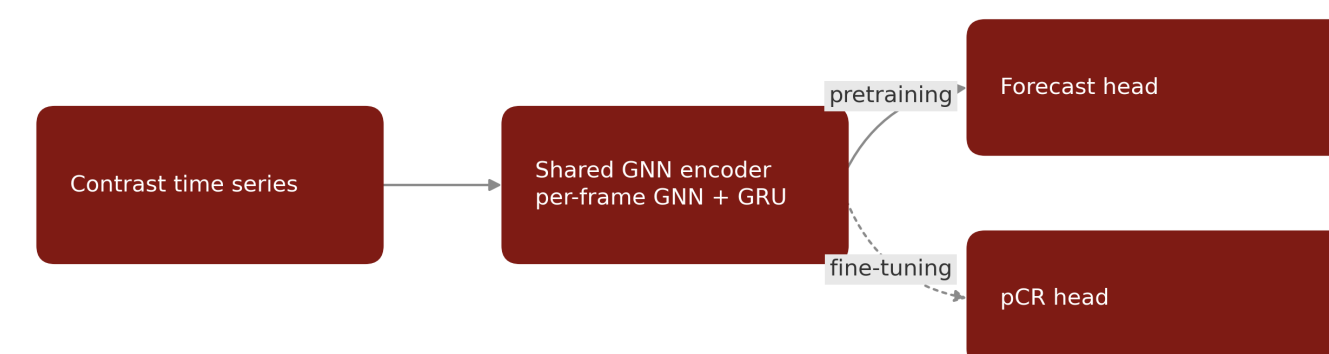
Node features (7). Peak time • peak enhancement • time to enhancement • wash-in slope • wash-out slope • positive AUC • vessel radius. **Edge features** summarise each segment’s geometry and mean kinetics.

What is thrown away. Each node’s 24-point enhancement curve becomes six scalars before the model sees it, and edges carry no flow direction — so the graph can only average static per-node kinetics.

Contrast-forecasting pretraining

A new contribution of this project.

The models here never see contrast *move*. So we pretrain the encoder on a label-free task: **given the first frames, predict every node’s enhancement over the next frames**, using message passing — so the encoder has to learn how the bolus propagates along vessels. After Tang et al. [2], who forecast EEG.



Needing no labels is the point: pretraining can use $\approx 1,000$ unlabelled ultrafast graphs, against 179 labelled ones. Built and validated end to end across all three graph representations.

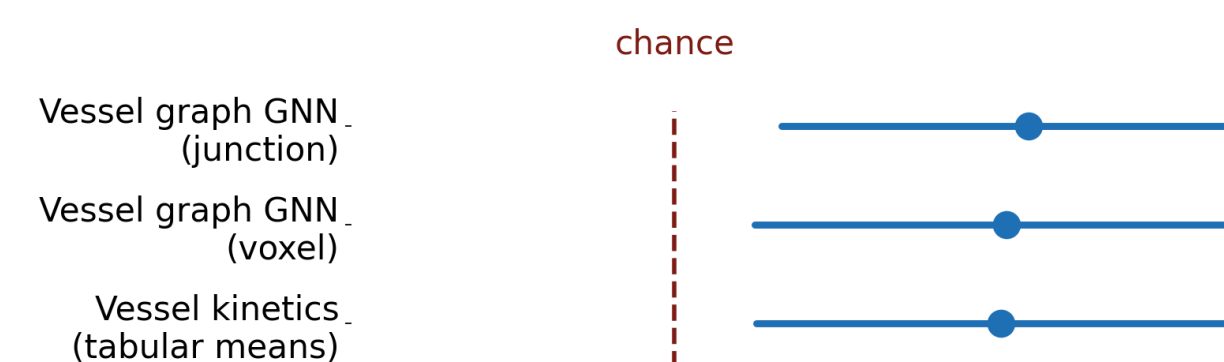
Pre-registered gates. Beat trivial forecasters, then a graph-free forecaster, then random init on pCR by $>2\times$ the seed spread.

TBD Gate results on ultrafast data.

Forecasting MAE against the trivial and graph-free baselines, and pretrained-vs-random pCR AUC over ≥ 5 seeds.

1 Signal in the vessels? **YES**

UChicago ultrafast DCE-MRI, $N = 179$, **vessels only** — no tumour mask, no clinical variables. Pooled out-of-fold AUC, case-resampling bootstrap CIs.



2 Does the graph add? **NOT YET**

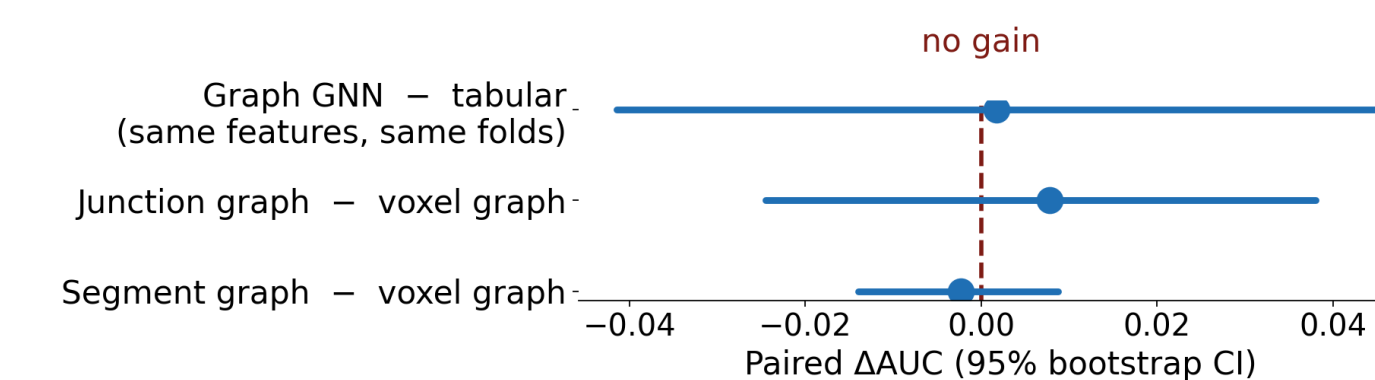


Fig. 3: Every paired interval contains zero.

Same features, same folds, paired bootstrap. The graph GNN does not beat a logistic regression on seven per-case feature means ($\Delta\text{AUC} +0.002$, 95% CI $[-0.041, +0.046]$), and no structural representation beats the voxel graph. At $N = 179$ that interval is ± 0.04 wide — a smaller real advantage could not be detected here.

TBD Sub-question 2 with temporal node representations.

Paired ΔAUC of a graph model that reads each node’s full enhancement curve, against the same tabular bar on the same folds.

Conclusions

- Vessel kinetics predict pCR **from vessels alone**.
- The signal is in the **features**, not the representation.
- Graph topology, as encoded today, is not earning its cost.

Next. Nodes get the curve, not six scalars; run the pretraining gates; add tumour features; grow the labelled cohort.

References and acknowledgments